CareCompass

Team Diligence | IIT Patna | Synthetic hackathon prototype | 24 October 2026

Problem and working prototype

Families describe care needs in English, Hindi or Hinglish. CareCompass routes a request to the supported care pathway, asks one decision-relevant question when needed, and ranks eligible synthetic caregivers. It supports editable voice transcripts and simulated booking. It does not diagnose, prescribe, dispatch or make real appointments.

API-first architecture

FastAPI/Pydantic validates the exact organiser response. The API, browser interface and batch exporter share one core: replay user evidence, screen locally for emergencies, optionally extract structured facts through Groq Account A, recheck mandatory safety, ask an approved question or route, then match caregivers. UI-only evidence and concern metadata are kept outside the scoring schema. A native mobile-first HTML/CSS/JavaScript interface is served by the same process, avoiding a Node build layer. It provides Hindi/English selection, Hinglish input, voice recording, transcript editing, follow-ups, ranking explanations, no-match states and simulated confirmation. Docker includes FFmpeg for media validation.

Two explicitly separate Groq accounts

Account A uses configurable openai/gpt-oss-20b structured extraction, with required fields and strict JSON schema. Account B uses configurable whisper-large-v3 transcription; whisper-large-v3-turbo is supported by the benchmark script. Separate server-side HTTP clients bind separate credentials. No keys enter browser code. Bounded retries, deadlines and malformed-output checks prevent silent provider failures. With no text key, a clearly labelled local rules baseline runs. Without an audio key, voice reports unavailability. Live Groq calls and comparative Hindi/Hinglish transcription quality have not been verified. Provider tests use labelled transport stubs; this is not a claim of live integration success.

Adaptive state and explainability

State retains patient identity, age, duration, temperature/unit/timing, associated symptoms, warning-sign statuses and evidence references. It interprets short answers using the preceding question, accepts corrections and separates patients. A controlled bank asks one relevant question per turn, normally caps at four, and never delays an established emergency. Explanations use supplied facts and computed roster information, without disease probabilities or invented lab values.

Safety policy and known contract gaps

Any affirmed active controlled warning sign forces emergency_112/emergency, no matches and no follow-up delay. Local detection runs before a network call and after extraction. Negation, historical and hypothetical statements are distinguished within finite supported patterns. A transcript warning is preliminary and corrections remain editable. Emergency actions say call 112; no dispatch is implied. Persistent/high fever surfaces prompt medical-assessment concern without claiming dengue or falling platelets. The taxonomy lacks urgent in-person physician assessment; same-day clinician consultation is an explicit coordination fallback, not equivalence. Infant fever uses a conservative organiser pathway: NHS guidance establishes urgent assessment, not ambulance necessity for every fever. Broader warning-sign coverage and clinical mentor review remain necessary.

Deterministic caregiver matching

Eligibility requires active status, compatible role, explicit procedural skills and home-visit radius. Availability uses an injectable IST reference clock. Haversine distance is ignored for virtual doctors. Ranking follows the README weights: language 0.35, proximity 0.30, skill 0.20 and rating 0.15. Future availability is unconfirmed and future booking is disabled. No insulin procedure match is offered because the roster lacks explicit subcutaneous eligibility.

Measured evaluation and data-quality limits

All 340 provided rows were inspected: only 66 literal messages, empty histories and six reasoning templates. Every emergency has the same two flags; 38 rows instead describe bleeding or unresponsiveness. Proposed corrections are stored separately; original data is unchanged. Urgency is fully determined by care type in the supplied labels. The rules baseline on all 340 examples measures care macro-F1 0.8698, urgency macro-F1 0.8736, emergency recall 56/56 and schema compliance 1.0. Matching label-agreement proxies are Hit@1 0.4976, Recall@3 0.5909 and NDCG@3 0.5452 on 209 nonempty-reference rows. A 103-row scenario-group-disjoint subset is also published. Rules were authored after inspecting templates: neither evaluation is blind hidden-test evidence or clinical validation. No official aggregate score is claimed. Tests cover all nine controlled emergency pathways, late escalation, negation, correction, patient isolation, procedural eligibility, transport/provider errors, separate credentials and FFmpeg decoding/silence. Browser visual QA, real microphone sessions, live Groq calls and Docker execution remain outstanding in this environment.

Sources, attribution and disclosure

Sanocare synthetic care-triage dataset v1 (2026), supplied for IIIT-Delhi. FastAPI, Pydantic, Uvicorn, HTTPX, Starlette, FFmpeg and pytest; Groq-hosted GPT-OSS and Whisper families. OpenAI ChatGPT assisted code, analysis, tests and documentation. Official sources: console.groq.com/docs/structured-outputs; console.groq.com/docs/speech-to-text; nhs.uk/symptoms/shortness-of-breath; nhs.uk/symptoms/fever-in-adults; nhs.uk/symptoms/fever-in-children; india.gov.in/directory/helpline. Full URLs and policy assumptions are in docs/ARCHITECTURE_AND_POLICY.md. Synthetic use only; no production or clinical readiness claim.